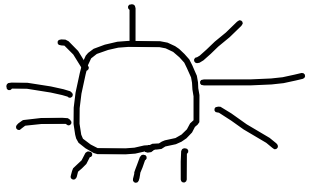
Energy from the Sun



PV- pedia

Nuclear Fusion



A reaction in which two light atomic nuclei combine to form a single heavier one while releasing massive amounts of energy.

Kelvin



Direct Current



Renewable Energy



Convection



Nuclear Fusion



A unit of measurement for temperature.

Kelvin



Direct Current



Renewable Energy



Convection



Nuclear Fusion



Kelvin



Direct Current



Electrical current which flows consistently in one direction.

Renewable Energy



Convection



Energy derived from natural sources that are replenished at a higher rate than they are consumed

Nuclear Fusion



Kelvin



Direct Current

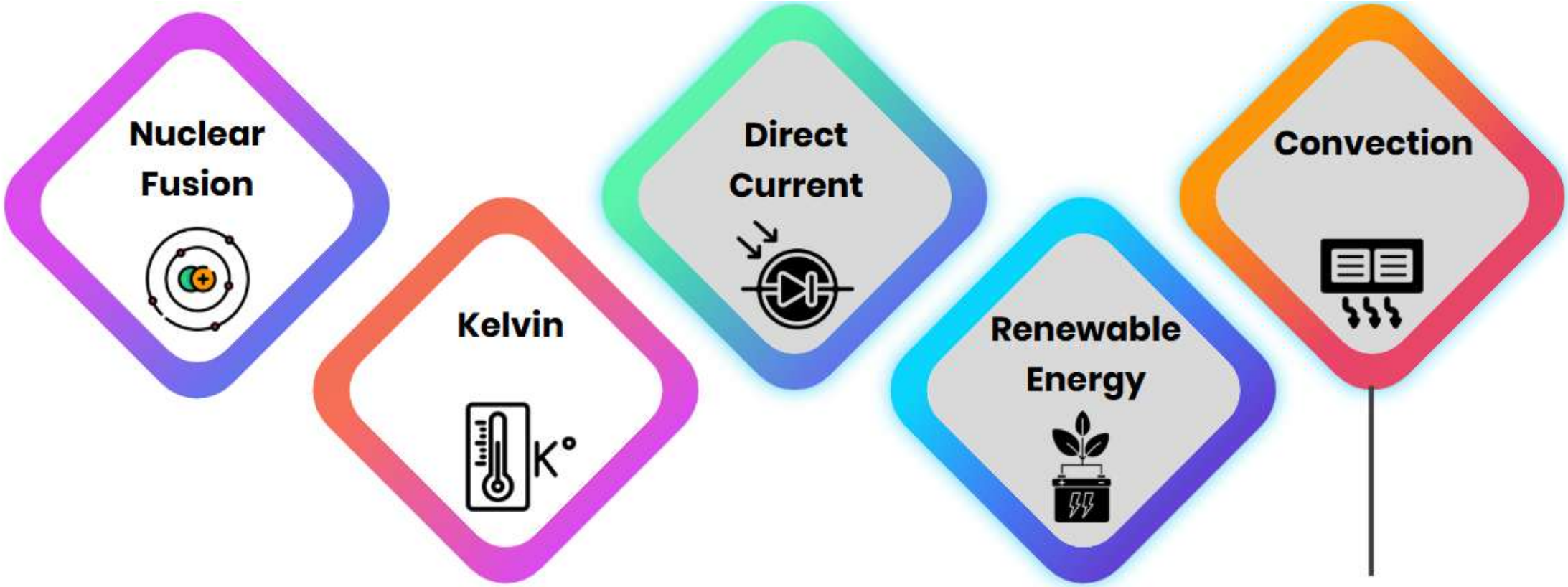


Renewable Energy



Convection





Transfer of heat through movement of fluids(liquids or gases). It occurs when a fluid is heated, causing it to expand and decrease in density. The less dense, warmer fluid rises, while the cooler, denser fluid sinks. This movement creates a circulation pattern that facilitates the transfer of heat.

2.2.1: Current

What is current?

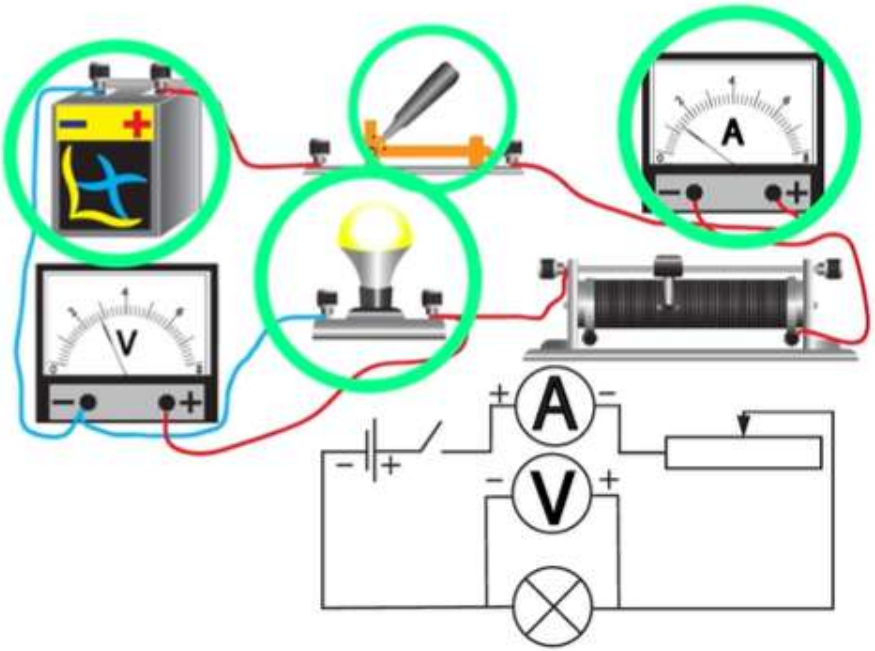
- The amount of charge flowing through a particular area in a unit time is electric current.

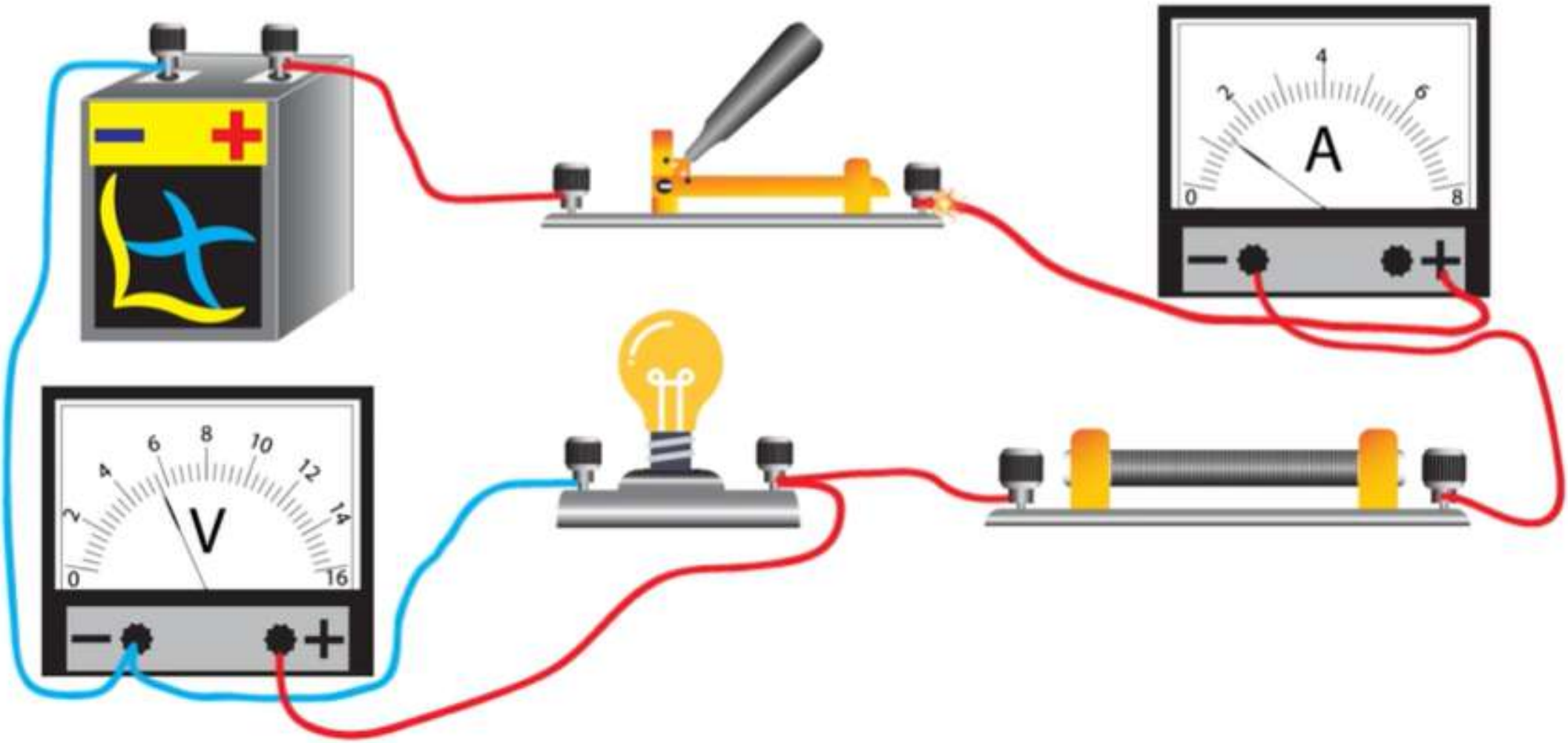
- Current is the rate of flow of electric charges.



Understanding the definition of Current through LED Lamp

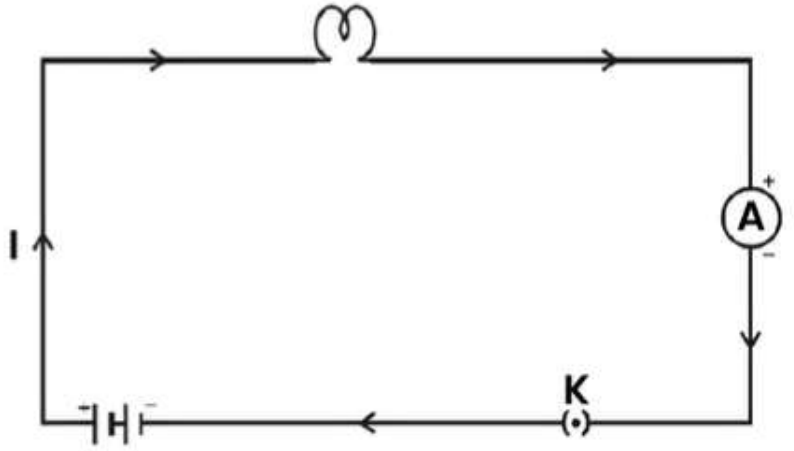
An electric circuit comprising –
cell , LED Bulb , ammeter and switch




A
0
2
4
6
8
V
0
2
4
6
8
10
12
14
16

Electric current unit - “Ampere”





2.2.2 Direct Current (DC)and Alternating Current (AC)

Direct current (DC)

Refers to the unidirectional flow of electric charge.

Electric current flows in one constant direction through a conductor, as opposed to alternating current (AC) which frequently reverses direction.

Alternating Current (AC)

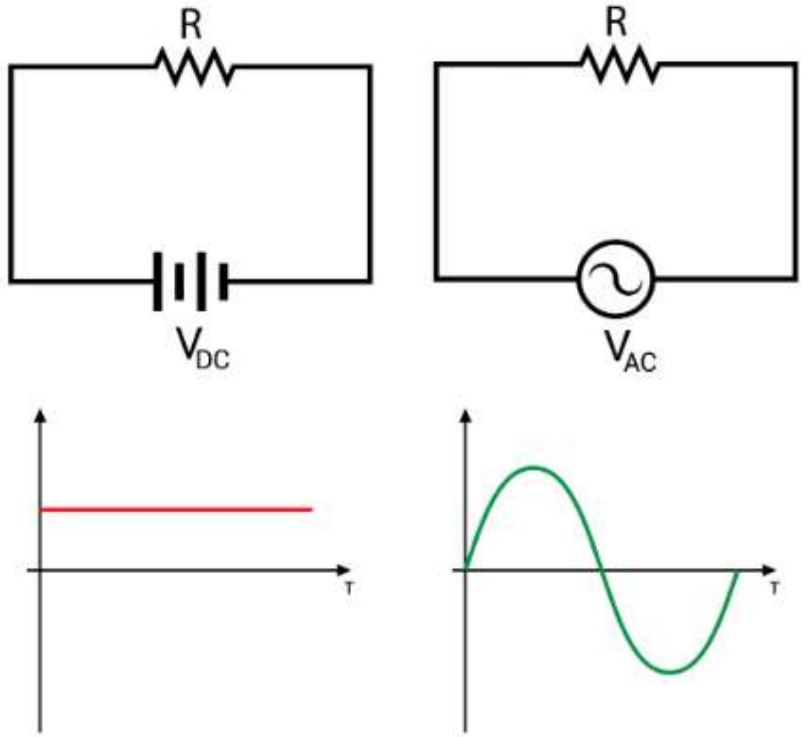
It is a type of electric current where the direction of flow of electrons reverses itself periodically.

Unlike Direct Current (DC) which flows constantly in one direction, AC constantly changes direction, creating a cycle.

2.2.2 Direct Current and Alternating Current

AC is characterized by frequency.

India	**50 Hertz (cycles per second)**
USA and Canada	**60 Hertz (cycles per second**



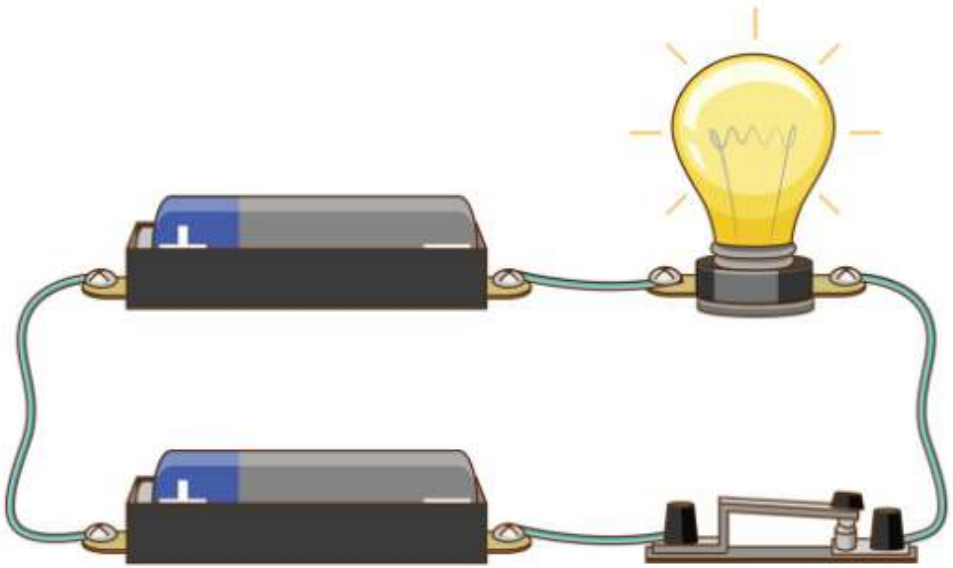
Alternating Current and Direct Current - Circuit and Waveforms

2.2.3 Voltage: Electric Potential and Differential Potential

What makes electric charge flow?

Electric charge flows due to a difference in electrical pressure, or potential difference (voltage), between two points.

Potential difference is the work done to move a unit charge from one point to another and is measured in volts (V).



2.2.4 Single Phase and Three Phase Voltage

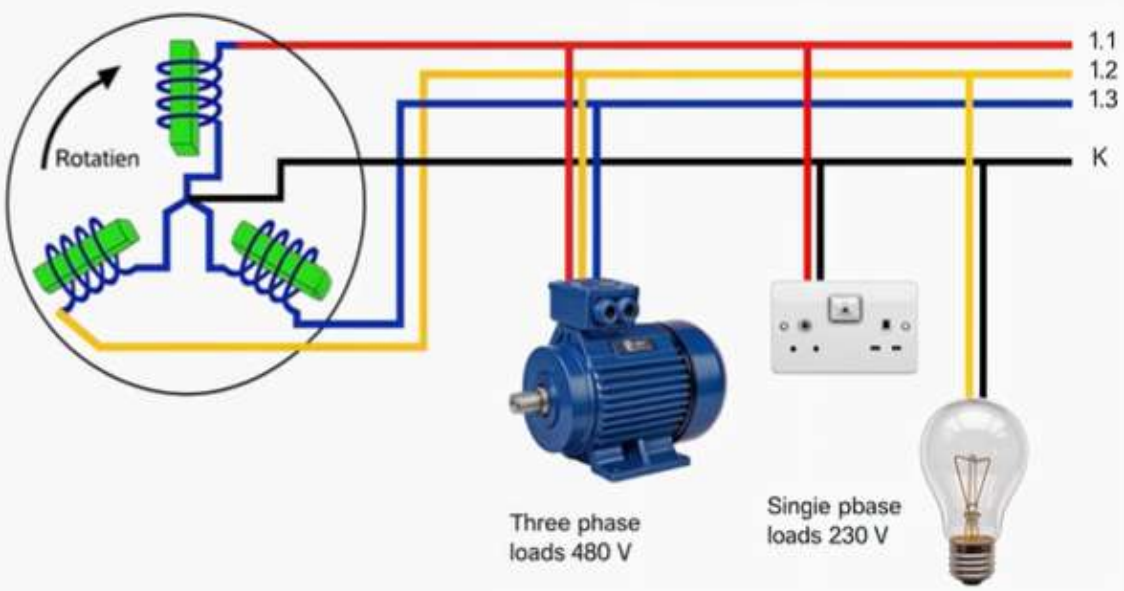


Single-phase voltage refers to a system where voltage is supplied through a single alternating current (AC) waveform.

Three-phase voltage, on the other hand, involves three alternating currents with equal voltages and frequencies.

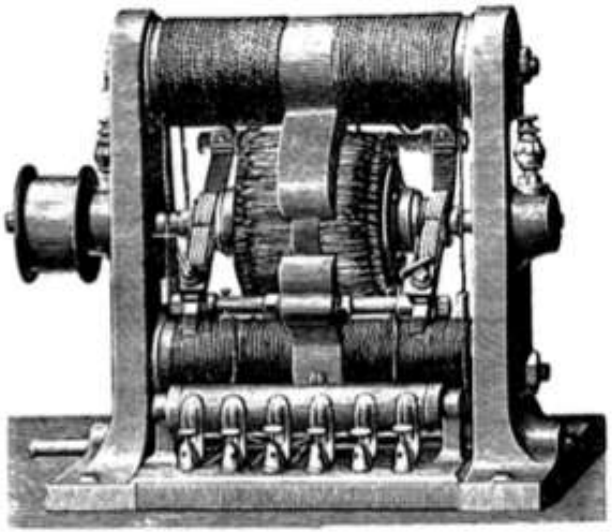


Single –Phase and Three-Phase Voltage

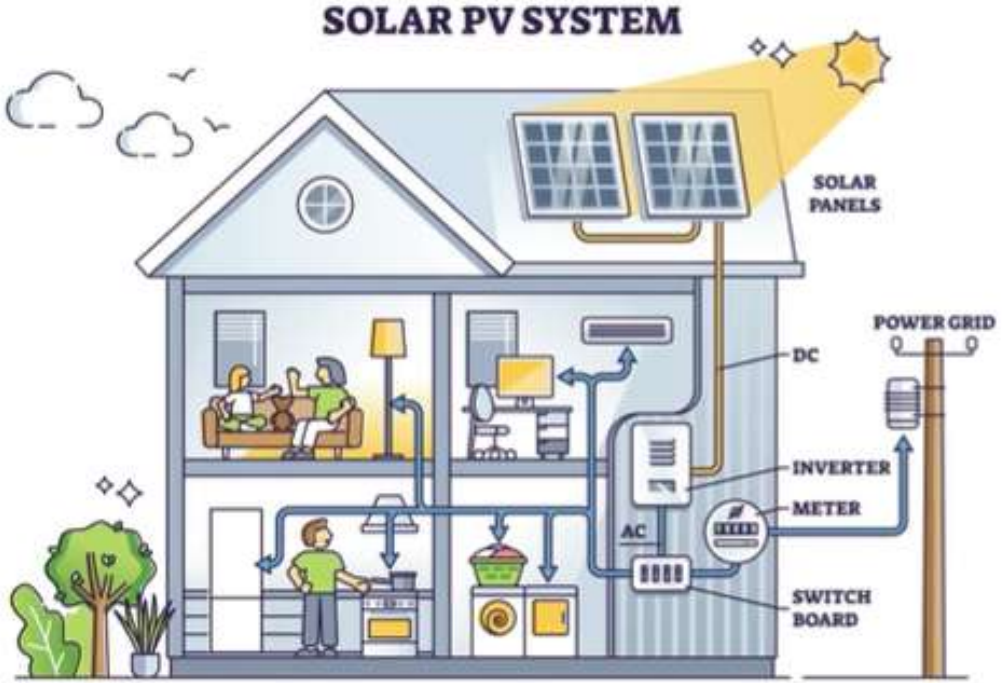


DC and AC Voltage

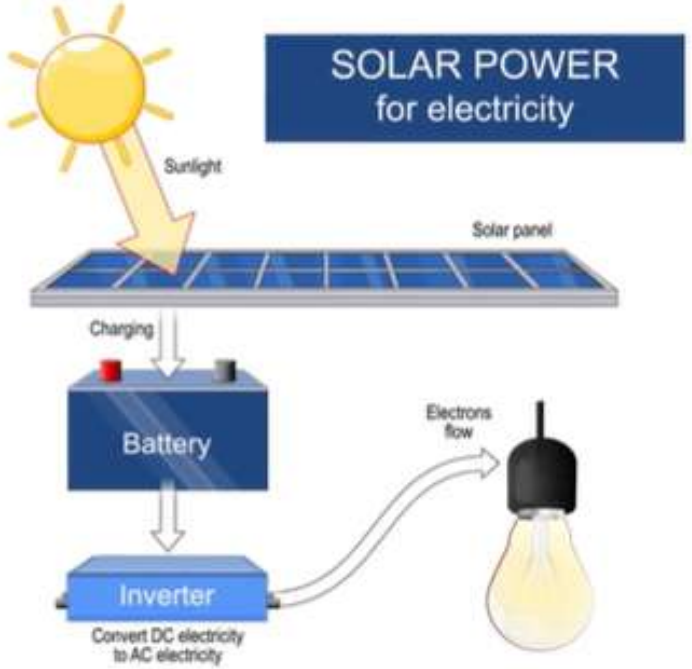
- If the voltage flow gives rise to DC flow, it is called DC Voltage.
- If the voltage source gives rise to the flow of AC form of current in a circuit is called AC voltage.
- AC became more popular in the 19th century because it was cost-effective.



A 19th Century illustration of the Ferranti-Thompson dynamo, a generator makes AC electricity (AC Alternating Current) using electromagnetism.



Solar panels produce DC electricity. To be compatible with the power grid that uses AC, this DC needs to be converted to AC.



An “inverter” is a device that performs this DC to AC conversion for integrating solar power into the grid.

Single Phase Supply: A two-wire Alternating Current (AC) power circuit.

Two types of wires are:

- Live Wire
- Neutral Wire

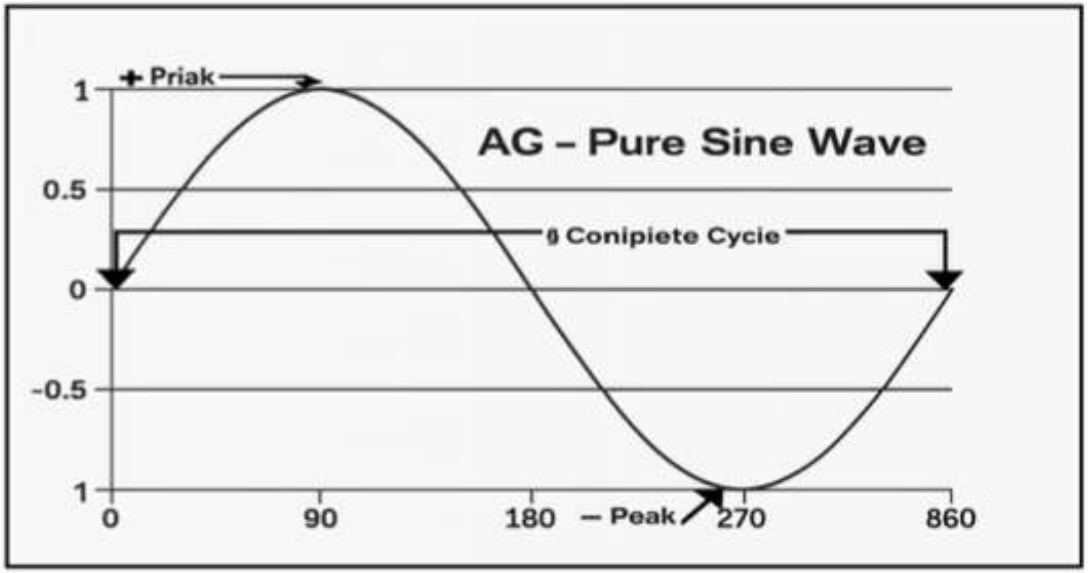


In India, 230V is the standard single-phase voltage with one 230V live wire and one neutral wire.



Electrical appliances are connected between the live and neutral wires for them to function.



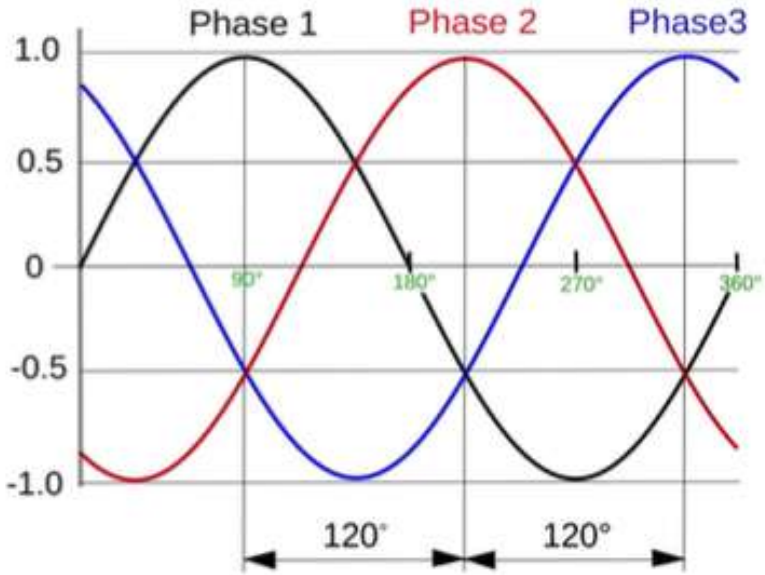


Three-Phase Supply: A three-wire Alternating Current (AC) power circuit.

There are types of power wires:

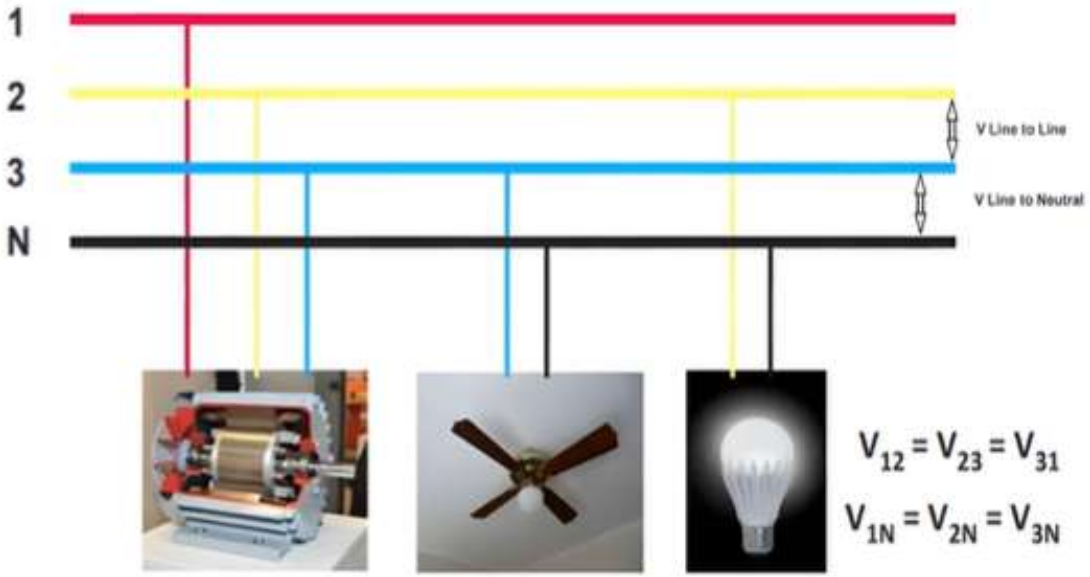
- Phase 1
- Phase 2
- Phase 3
- Neutral Wire

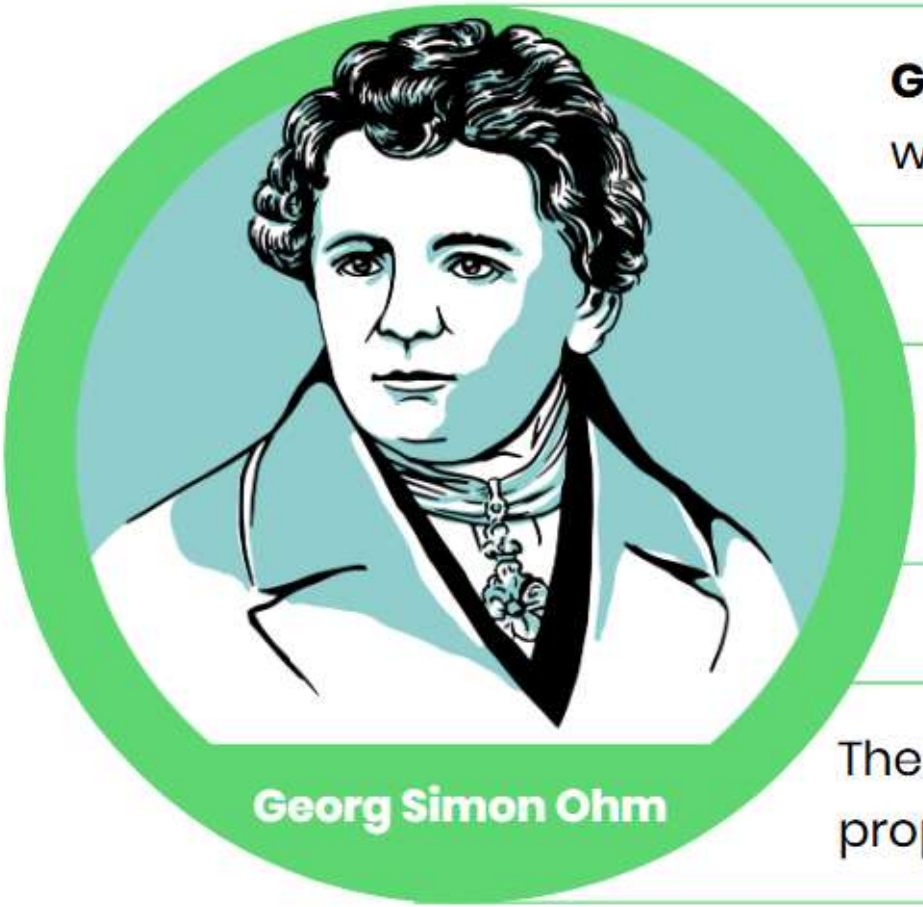
The phase order is 1–2–3. This sequence repeats each cycle, and sets the frequency of the power system.



3 Phase power arrangement:

- Saves on electrical construction costs.
- By reducing the current requirements.
- Required wire size.
- The size of associated electrical devices.
- Reduces energy costs.





Georg Simon Ohm

Georg Simon Ohm, a German physicist was the first to verify Ohm's Law.

Relationship between electric current and potential difference.

The current that flows is directly proportional to the voltage applied to it.

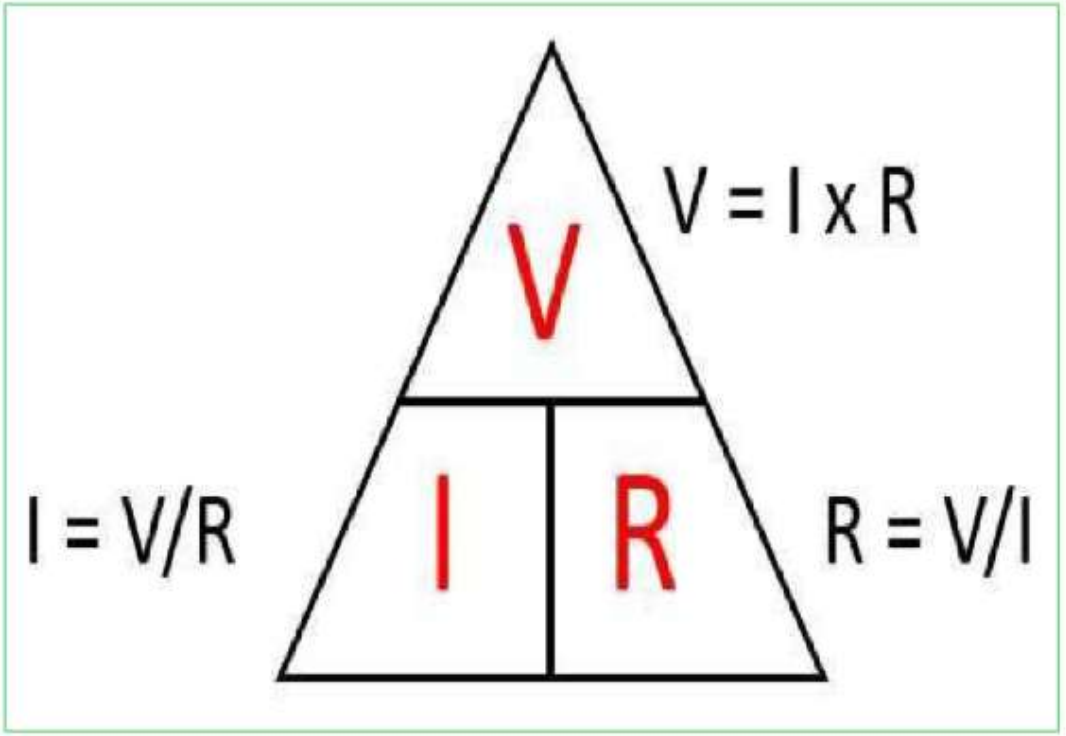
Mathematically, Ohm's Law is expressed as:

$$V=I\times R$$

Where:

- V is the voltage across the conductor in volts (V).
- I is the current flowing through the conductor in amperes (A).
- R is the resistance of the conductor in ohms (Ω).

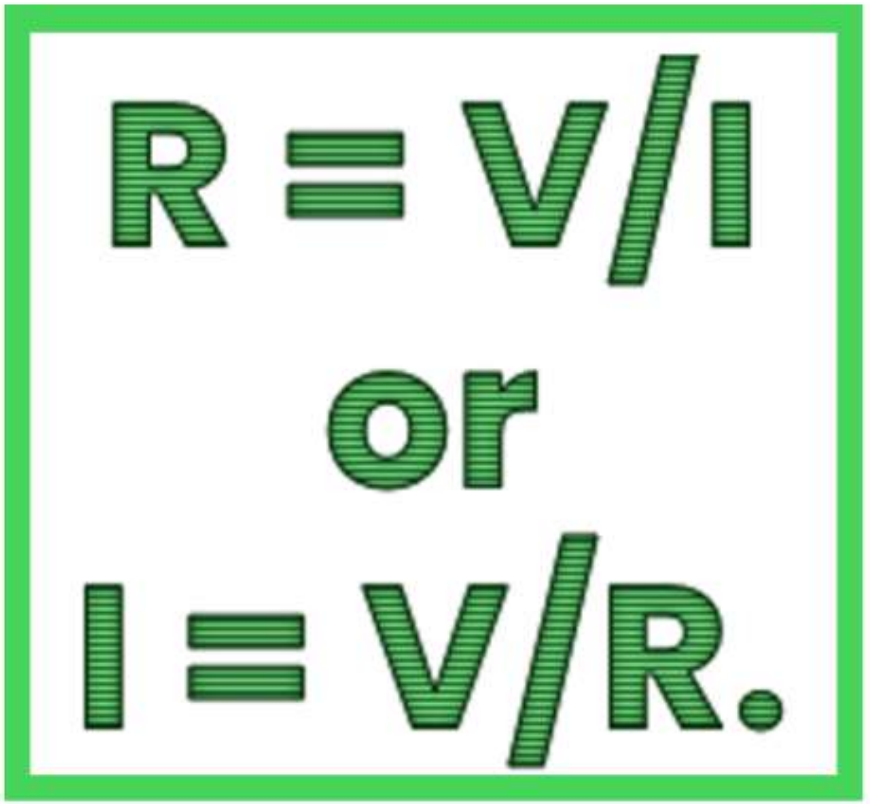
The temperature and the other physical factors remain constant.



2.2.6 Resistance

Understanding Resistance

- Resistance is the material property that opposes the flow of electricity.
- Current is inversely proportional to resistance.
- Doubling the resistance halves the current.



What is Resistivity?

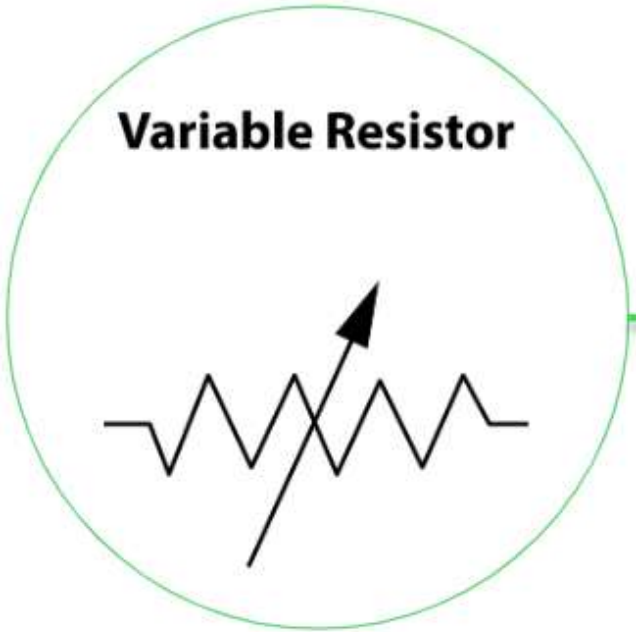
- Resistivity is the material property that reflects its inherent ability to resist electric current.
- Denoted by "ρ"
- Mathematical Representation of resistance in terms of resistivity:

$$R = \rho l / A$$

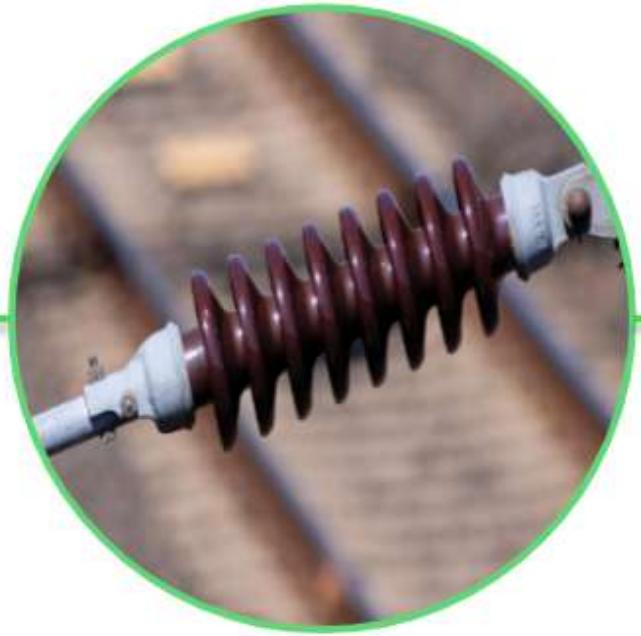
Here,

"l" denotes the length of the conductor
"a" denotes the cross-sectional area of the conductor

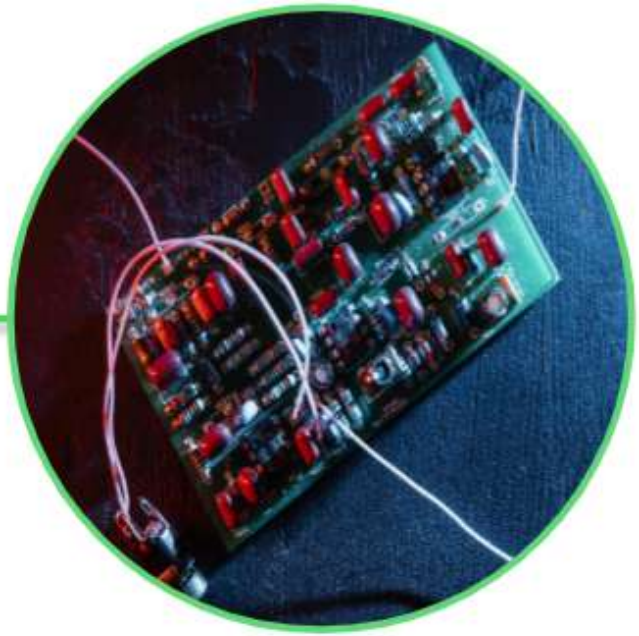
- The equation conveys that the resistance of a conductor depends on three factors which are Resistivity, Length, and Cross-sectional area.



These components adjust the current in a circuit without affecting the voltage source.



High-resistance material hinder current flow and are insulators.



Materials with low resistance allow easy current flow and are conductors.

Points to remember:

- A component of a given size that offers a low resistance is a good conductor.
- A conductor having some appreciable resistance is called a resistor.
- A component of identical size that offers a higher resistance is a poor conductor. An insulator of the same size offers even higher resistance.



2.3.1 Combination of Resistors

This table here compares Voltage & Current in Series v/s Parallel connection

For 'n' Lamps Connected across a Source	Series	Parallel
Voltage	Divided as per Resistance of Individual Loads	Same across all Loads
Current	Same flowing through all Loads	Divided as per Resistance of Individual Loads

Series

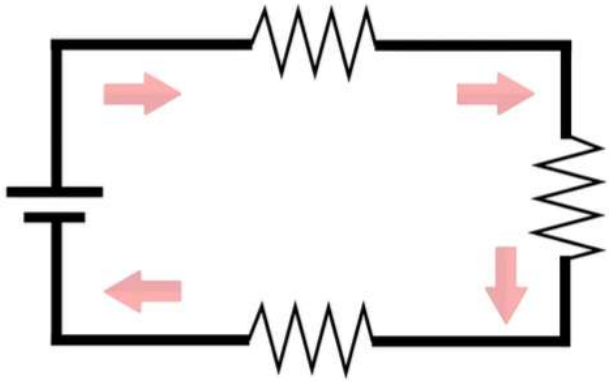
- In a series, the current is the same in every part of the circuit.

$$I_t = I_1 = I_2 = I_3 \ \ldots = I_n$$

- Total resistance is the sum of individual resistances.

$$R_t = R_1 + R_2 + R_3 \ldots R_n$$

- Total voltage equals the sum of individu



Calculation of Individual Voltage,

Add up the resistance of all the individual resistors

Total Resistance (R_t)= R_1+R_2+R_3+R_n

Calculate Current flowing through the Circuit,

$I_t = V_s / R_t$

Also, It =I_1 =I_2 =I_3 =...=

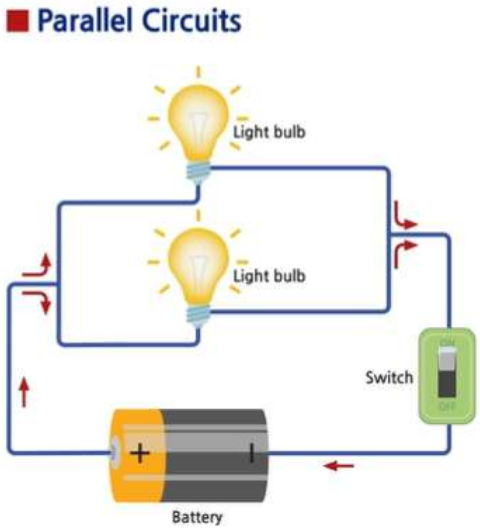
Calculate voltage through each element/LED lamp:

V_n= I_n x R_n

Voltage is added in a series branch	Vt= V_1+ V_2+V_3+...+V_n
Current remains same across the series branch	I_t=I_2=I_3=I_4= I_n

Parallel

- In a parallel circuit, components are connected across the same voltage source with two separate branches.
- It offers multiple wider paths for current to flow.
- Current through each element is calculated individually.



Calculation of Individual Current through each Element

Calculation of Total Resistance,

$1/R_t = 1/R_1 + 1/R_2 + 1/R_3 + \ldots 1/R_n$

Current is added in a parallel branch	$I_t = I_1 + I_2 + I_3 + \ldots + I_n$
Voltage remains same across the parallel branch	$V_t = V_1 = V_2 = V_3 \ldots = V_n$

Voltage remains same and is represented as,

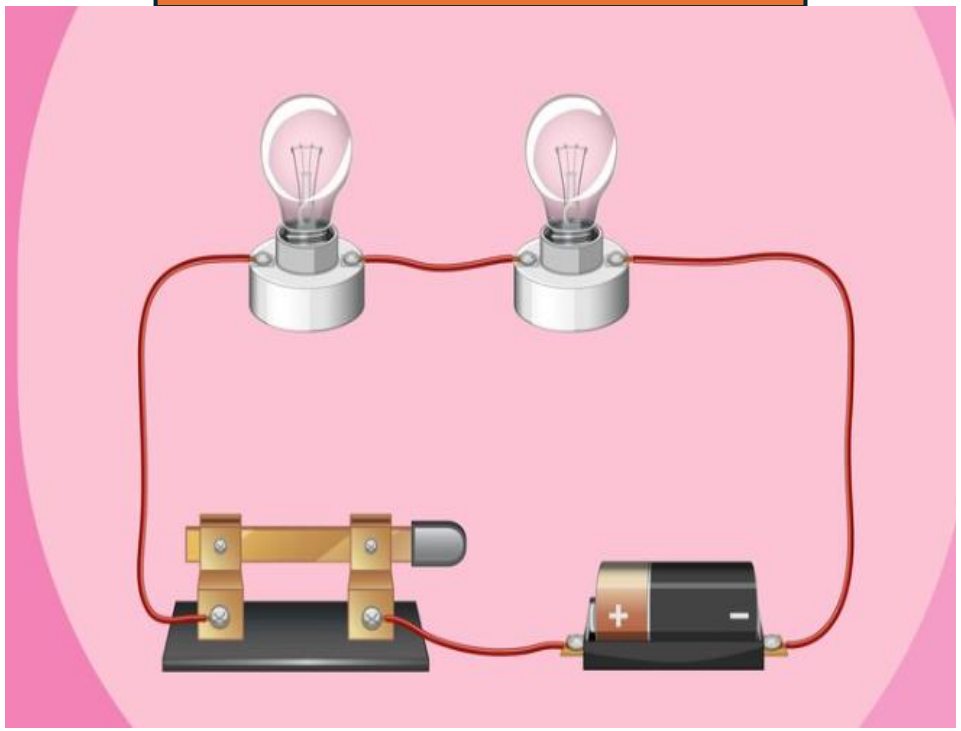
Vs=V1=V2=V3=...=Vn

Calculation of current through individual element

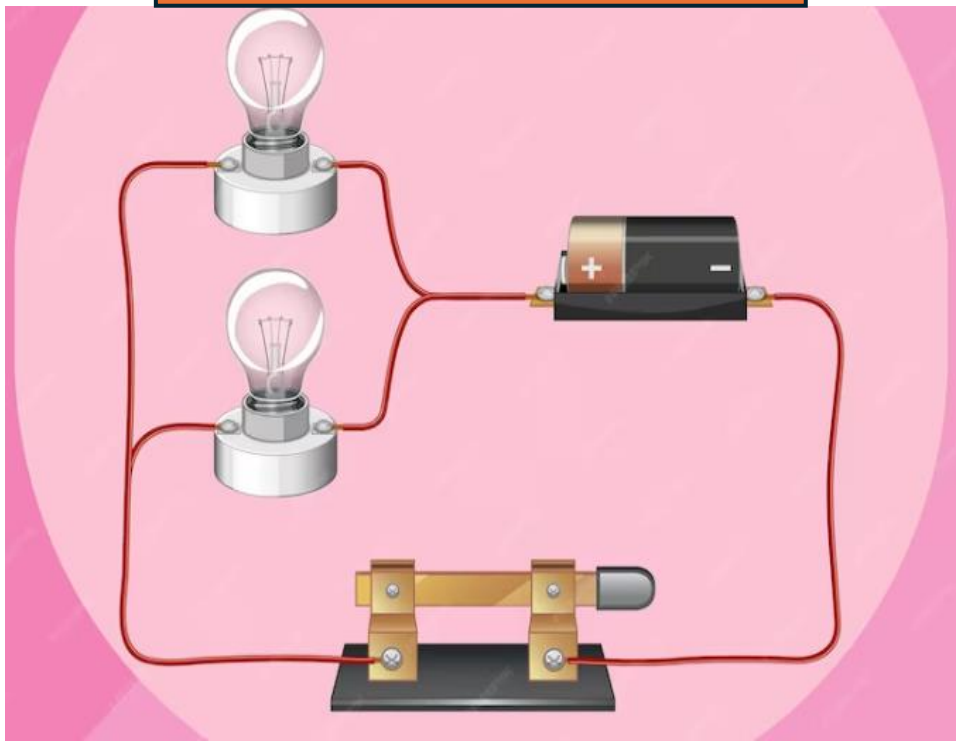
In=Vs/ Rn

Difference between Series and Parallel

Series



Parallel



Series

- The same current (I) flows through all components in the circuit.
- The voltage source (Vs) experiences a voltage drop across each component in the series, and the total voltage (Vt) equals the sum of these individual voltage drops (Vd). (Vt = Vd1 + Vd2 + ... + Vdn)

Parallel

- The current is divided and flows through each branch independently.
- The voltage across each branch (Vb) is the same as the voltage of the source (Vs).. (Vb1 = Vb2 = Vb3 = ... = Vs)

Series

The total resistance (Rt) of a series circuit is the sum of the resistances (R) of all individual components connected in series. (Rt = R1 + R2 + R3 + ...)

If one component in circuit fails, the entire circuit breaks because the current can no longer flow through a complete loop.

Parallel

The total resistance (Rt) of a parallel circuit is less than the resistance of any individual branch. (1/R1 + 1/R2 + 1/R3 + ...). (Generally, Rt < R1 and Rt < R2 and so on)

If one component in circuit fails, only that specific branch is affected. The current continues to flow through other branches.

Series

Used when all components need the same current or the total resistance needs to be high.

Examples: Series connection of LEDs with a current-limiting resistor, Christmas light

Parallel

Used when all components need the same current or the total resistance needs to be high.

Examples: Multiple appliances connected to a power strip, parallel resistors for voltage dividers.

Conclusion

Feature	Series Circuit	Parallel Circuit
Current Flow	Same through all components	Divided among branches
Voltage	Sum of voltage drops	Same as source voltage
Resistance	Sum of individual resistances	Less than any branch Ri
Component Failure	Entire circuit breaks	Only affected branch stops
Applications	Same current, high Rt	Independent control, redundancy

2.4: Measuring instruments

PV-pedia



Capacitance

Capacitance is the ability of a material object or device to store electrical energy in the form of an electric charge.

Continuity

Red Probe

Frequency

PV-pedia



Capacitance

Continuity

Continuity refers to a specific test function used to check for a complete electrical path between two points in a circuit. It essentially determines whether there is a low resistance path for current to flow.

Red Probe

Frequency

PV-pedia



Capacitance

Continuity

Red Probe

The red probe is conventionally used to connect to the positive terminal or point of higher voltage in a circuit when measuring voltage or resistance. It acts as the input for these measurements. It is used in Digital multimeter.

Frequency

PV-pedia



Capacitance

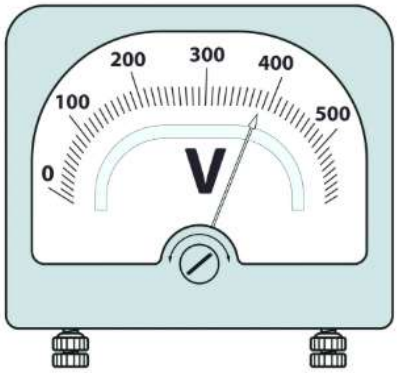
Continuity

Red Probe

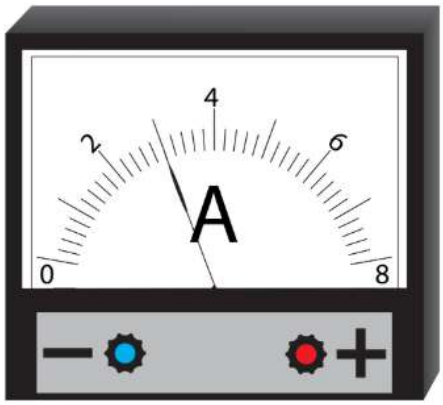
Frequency

Frequency refers to the number of repetitions of an event (like a wave cycle) per unit of time (usually seconds). It's important in various applications like AC (alternating current) circuits, signal analysis, and radio waves.

2.4.1 Voltmeter and Ammeter



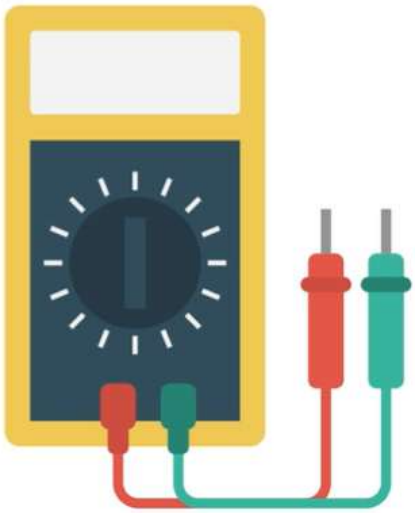
Voltmeter



Ammeter

Voltmeter

- Measures voltage in an electrical circuit.
- Measures the potential between any two nodes in a circuit.
- It is designed to have very high resistance.
- The range of the DC Voltmeter is 0-5 V, 0-10V , 0-20V etc.
- It is connected in parallel across the circuit elements where Voltage has to be measured.



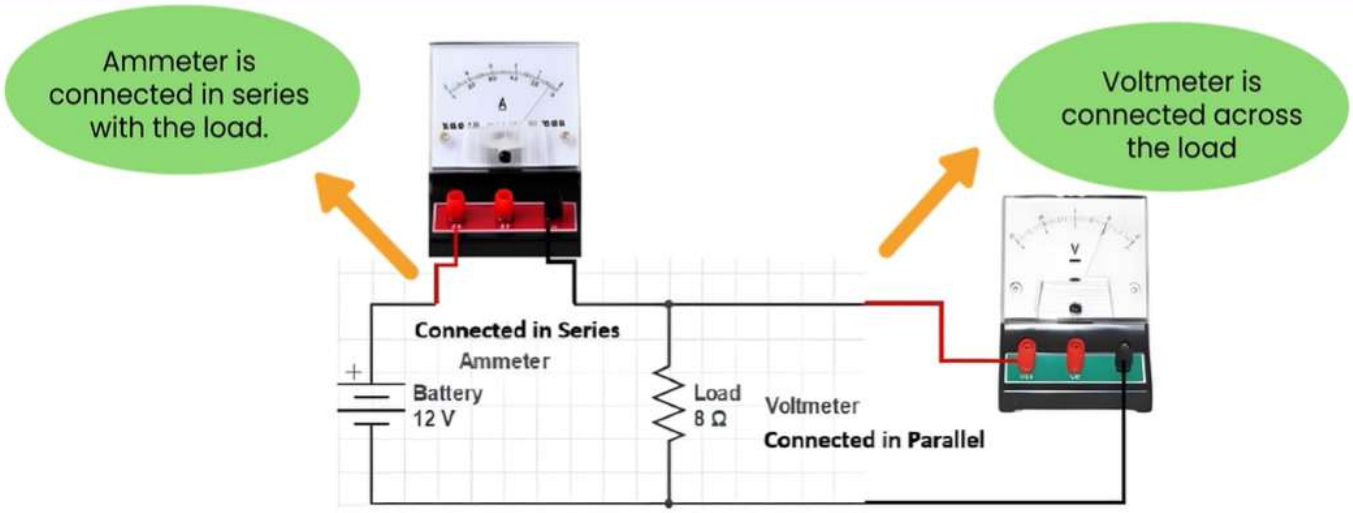
Ammeter

- Ammeter measures current in any branch of the electric circuit.
- The typical current range is from 0-1 A to 0-5 A for laboratory measurements.
- It is connected in series because the current in all elements remains the same.

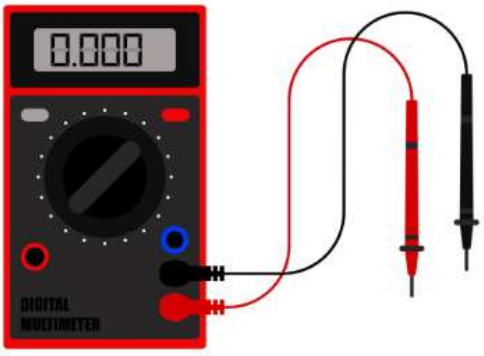


Thus, the Ammeter has low resistance. → Creates nearly zero voltage drop → Causing negligible interference in the circuit.

Voltmeter and Ammeter in Electric Circuit



2.4.2 Digital Multimeter and Clamp Meter



Digital Multimeter



Clamp Meter

Digital Multimeter

A versatile electronic device that can measure voltage, current, and resistance, with additional functions like temperature and frequency.



Clear readings

Measured value

Display Screen

Selection knob

Selection for measurement:
Voltage/Resistance/Current

mAV port

special port

Ports

Connecting probes

COM port

Clamp Meter

- A clamp meter is an electrical tester that combines a basic digitalmultimeter with a current sensor.
- It measures: AC current, AC and DC voltage, resistance, continuity, DC currents, capacitance, temperature, frequency and more.
- Measures to the nearest tenth of a unit (0.1 amps, for example).
- Safer to measure electrical parameters because no direct interaction with the bare electrical wire is required.





2.5: Power and Energy

2.5.1 Electrical Power

What is Electrical Power?

- Power is the rate at which you can do work.
- The conversion of energy from sunlight into electricity, either directly using photovoltaics or indirectly using concentrated solar power gives us solar power.



2.5.1 Electrical Power

What is Electrical Power?

- When electricity flows in an electrical circuit, it results into some work done.
- Power (P), is a measure of the speed or rate at which work is done.
- The S.I unit of Power is Watts (W).



2.5.2 Electrical Energy

Electrical energy represents the total work done

Energy consumed by an electrical appliance depends on two factors:

1. Power of an appliance
2. Duration of usage

The S.I unit of **Electrical Energy is Joule (J)**.



Mathematical Representation of Electric Energy

Electrical Energy = Power X Time

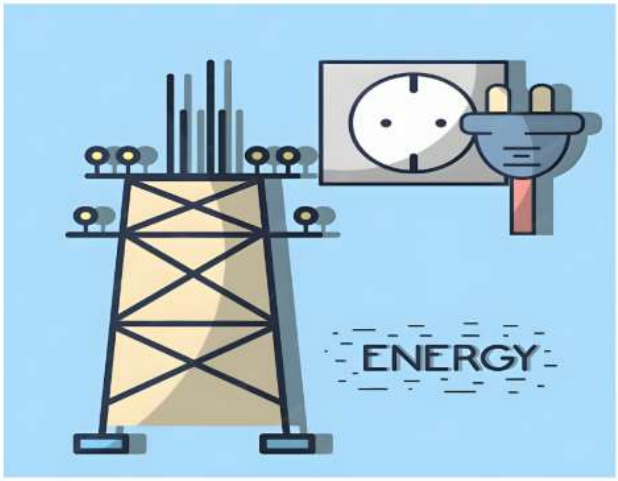
E(Wh) = P(W) X T(h)

When electrical energy changes with time – unit of energy is Watt-hour or Wh, and 1KWh = 1000 Wh

When electrical energy is a definite amount, then S.I. unit is Joules or J

where

1 (Wh) = 3600 Joules



2.6: Earthing and Lightning Protection

PV- pedia

Lightning	Over voltage	Electrocution	Equipotentially bonded	Surges	Downstream Electronic devices

Lightning is a giant spark of electricity in the atmosphere between clouds, the air, or the ground.



PV- pedia

Lightning	Over voltage	Electrocution	Equipotentially bonded	Surges	Downstream Electronic devices

Over voltage is the raising of voltage beyond the design limit of a circuit or circuit element.



PV- pedia

Lightning	Over voltage	Electrocution	Equipotentially bonded	Surges	Downstream Electronic devices

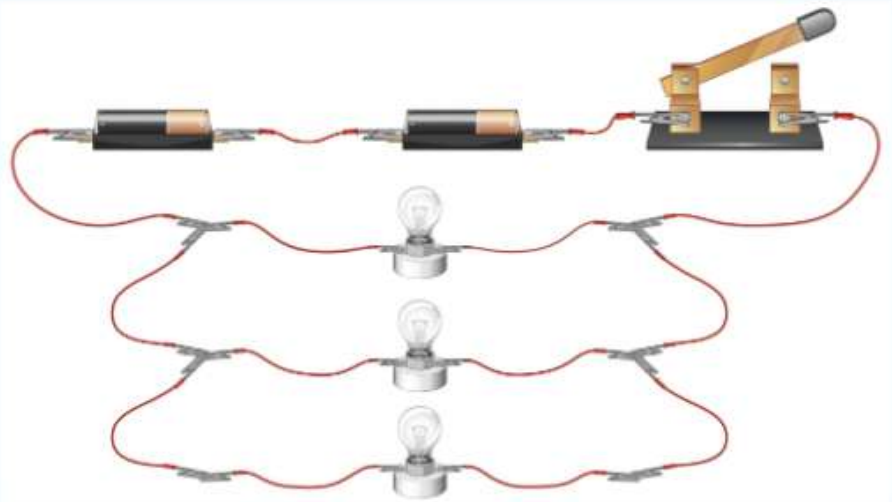
The injury or killing of someone through electric shock.



PV- pedia

Lightning	Over voltage	Electrocution	Equipotentially bonded	Surges	Downstream Electronic devices

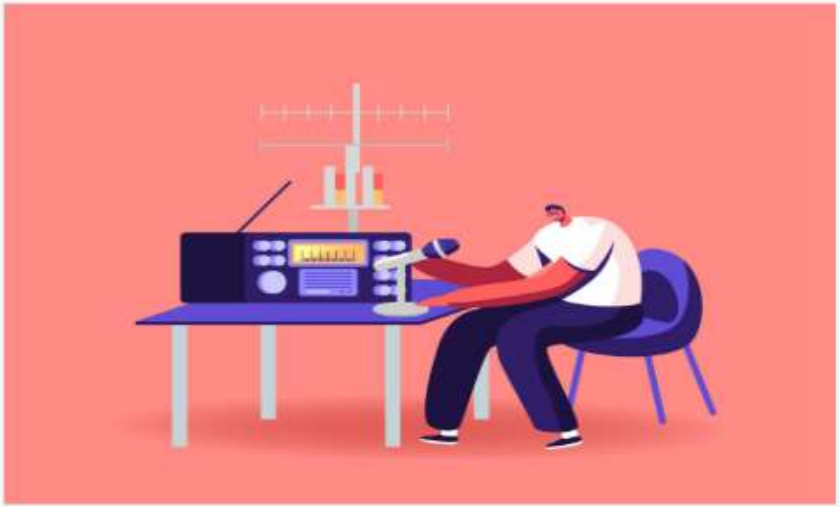
Equipotential bonding (EPB) is the process of electrically connecting metalwork and conductive parts, both exposed and extraneous, such that the voltage is the same throughout these various parts.



PV- pedia

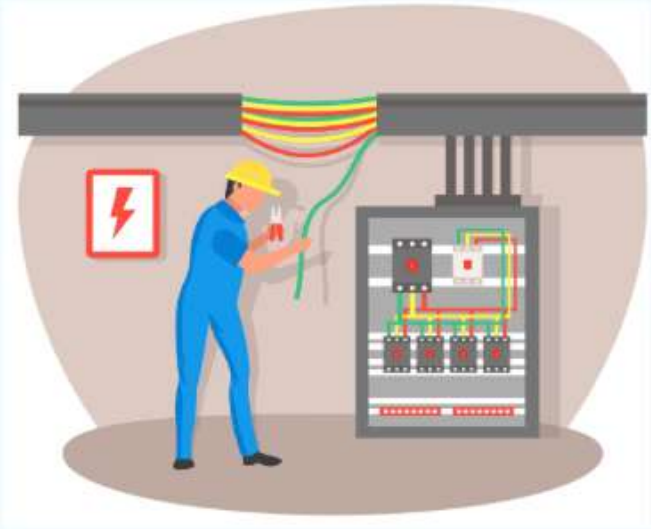
Lightning	Over voltage	Electrocution	Equipotentially bonded	Surges	Downstream Electronic devices

Surges, are brief overvoltage spikes or disturbances in a power waveform that can damage, degrade, or destroy electronic or electrical equipment.



Lightning	Over voltage	Electrocution	Equipotentially bonded	Surges	Downstream Electronic devices

Such devices have the circuit that is fed and controlled/ protected by the circuit breaker.



2.6.1 Lightning Protection System



Lightning Protection Systems

What?
Designed to safeguard structures and people from the damaging effects of lightning strikes.

Where?
Locations that are exposed to lightning

How?
They work by providing a safe path for lightning current to travel from the sky to ground, minimising potential harm.

When?
Installation during construction of the buildings or when the susceptibility of being directly affected by lightening increases.

Did you know ?

- PV systems do not increase the risk of buildings being struck by lightning.
- The PV generator must be connected to lightning protection systems.
- Suitable surge arrestors on the DC side in the panel junction box is recommended.
- Overvoltage protection on the AC side is also recommended.
- All potentially conductive systems like ACs, heating appliances, etc. must be connected with the earth.



Difference between External and Internal Lightning Protection

External Lightning

- External lightning protection includes all measures and devices for intercepting and providing an exit path for lightning.
- Consists of an interception unit, lightning wire, and earthing system.

Internal Lightning

- Internal Lightning protection includes safeguarding from the indirect effects of lightning.
- It has all the considerations and equipment like electronic devices switching gear.

- Shielded individual cables are recommended in a system that is exposed to the risk of lightning.



- The cross-section of the shield should be as per minimum standards.



- It is safe to arrange the shielded DC main cable along the building's side and to lead it towards the earth.

Surge Protection Devices

Surge arrestors are used to protect PV systems and downstream electronic devices.

Surge Protection Devices (SPD):
They are a combination of (MOV) and or/(GDT) which acts like a diode. They shunt the current created by the high voltage away from the protected sensitive areas whe



DC SPD



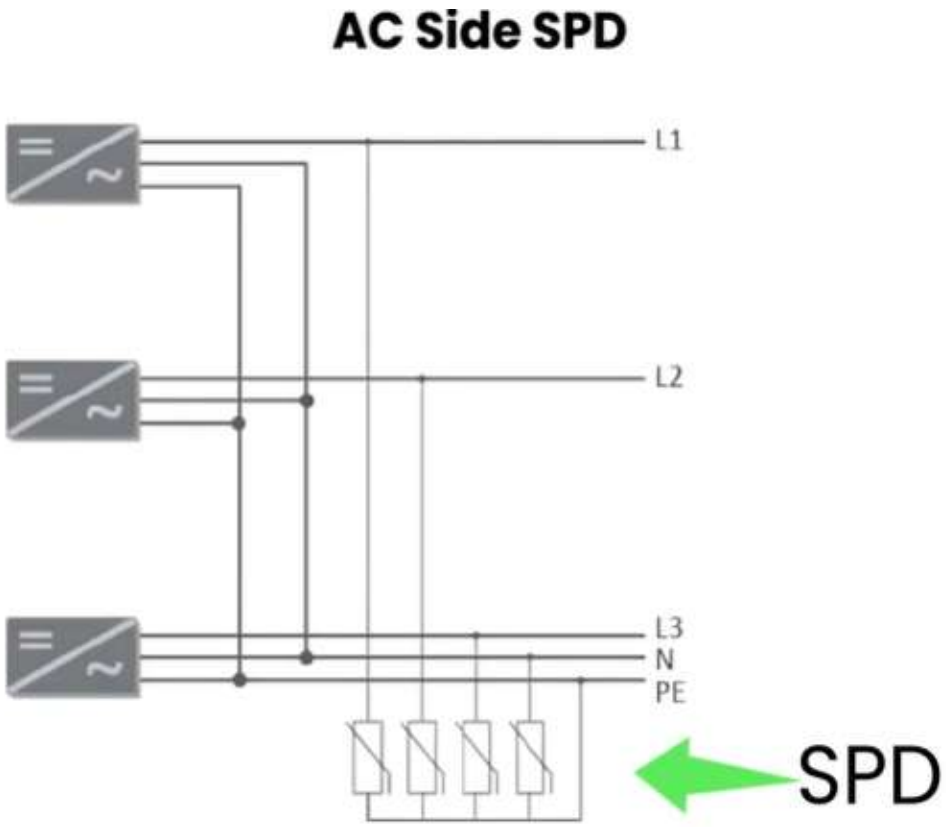
Single Phase AC SPD



Three- Phase AC SPD

Inverters can be connected to the same SPD if they are sharing the same grid connection.

The type of surge arrestors for the AC side are different. Surge protection should be according to the wiring specifications.



2.6.2 Earthing/Grounding

What is Earthing?

- It is the process of connecting the metallic (conductive) parts of an electrical system to the earth (ground) through a low-resistance wire.
- It plays a critical role in ensuring the safety and proper functioning of the electrical system.



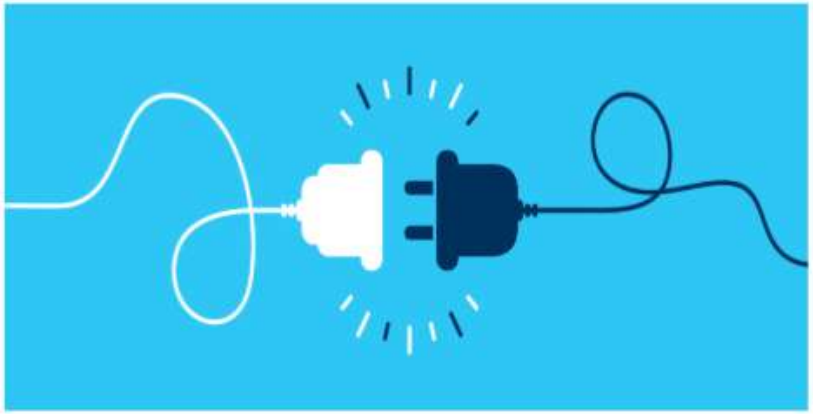
Safeguarding from Overvoltage

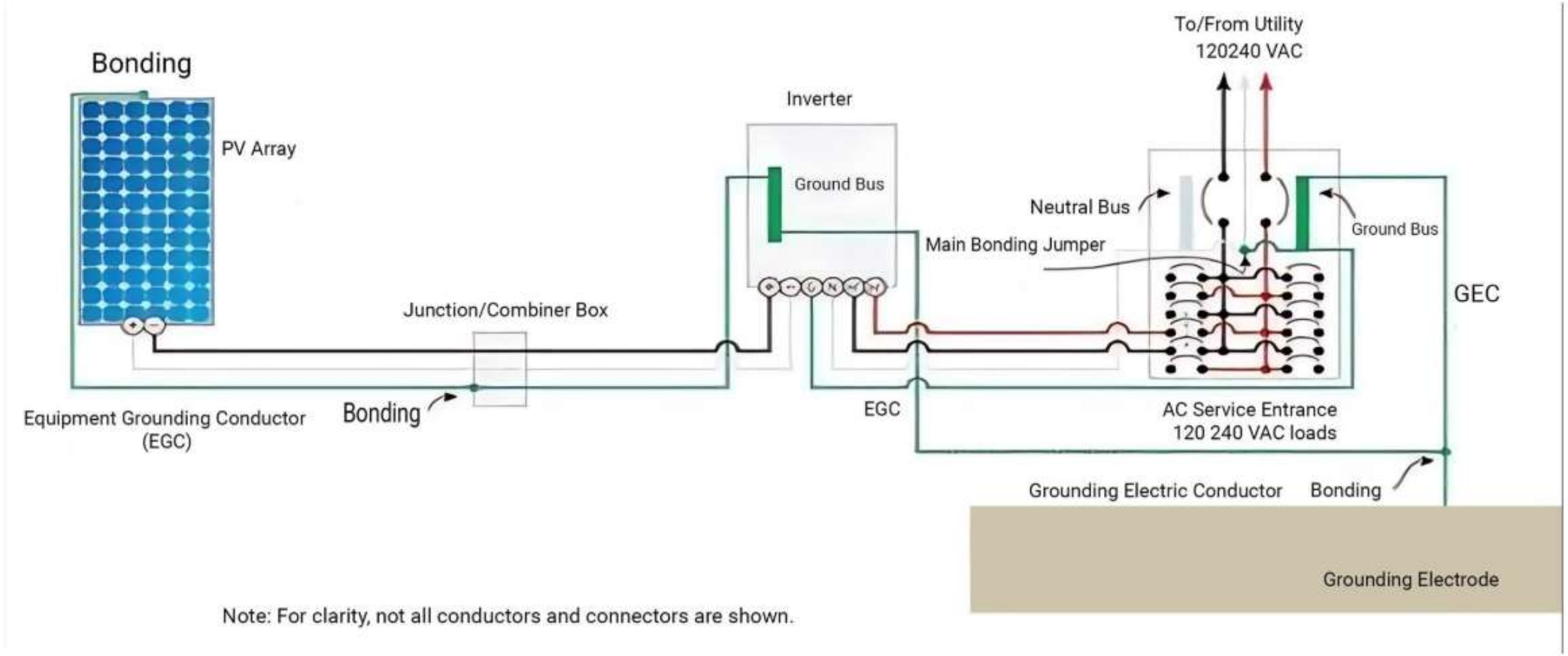
- The grounding conductor must be led to the earth along the shortest path possible.
- Loop formations must be avoided.
- Earthing conductors must be laid straight to ensure an optimum path.
- The mounting system of the PV panels must be equipotentially bonded.
- The material used for earthing/grounding conductors can include Copper (16 sq.mm), Aluminium (25 sq.mm), Steel (50 sq.mm)or other natural components.

Importance of Earthing

Earthing includes protection of:

1. Personnel: Engineers, technicians, and consumers from electrocution and fire.
2. Property: Equipment and facility from failure and fire.
3. Electric circuits from cable failures.





Note: For clarity, not all conductors and connectors are shown.